

Air University



Programming Fundamentals (Lab)

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Assignment # 04

Row (1-17)

- Libraries and Function prototypes.

```
1  #include<iostream>
2  #include<fstream>
3  #include<string>
4  using namespace std;
5
6  // function prototypes
7  void countWords();
8  void copyContents();
9  void longestWord();
10 void countLines();
11 void replacingWord();
12 void contentReversing();
13 void sortNumber();
14 void countVowels();
15 void findSpecificLine();
16 void mergeFiles();
17
```

Row (18-36)

- Main function.
- Displayed menu for each task.

```
18 int main() // main function
19 {
20     int choice;
21     do
22     {
23         cout << "\nMenu:" << endl;
24         cout << "1. Count Words" << endl;
25         cout << "2. Copy Contents" << endl;
26         cout << "3. Find Longest Word" << endl;
27         cout << "4. Count Lines" << endl;
28         cout << "5. Replace Word" << endl;
29         cout << "6. Reverse File Contents" << endl;
30         cout << "7. Sort Numbers" << endl;
31         cout << "8. Count Vowels" << endl;
32         cout << "9. Find Specific Line" << endl;
33         cout << "10. Merge Files" << endl;
34         cout << "0. Exit" << endl;
35         cout << "\nEnter your choice: ";
36         cin >> choice;
```

Row (38-69)

- Used a switch in main function and called every task's function which were created separately to perform every assigned task.

```
38 switch (choice) // handling user choice
39 {
40     case 1: countWords(); // counting words
41     break;
42     case 2: copyContents(); // copying contents
43     break;
44     case 3: longestWord(); // Longest word
45     break;
46     case 4: countLines(); // counting lines
47     break;
48     case 5: replacingWord(); // replacing word
49     break;
50     case 6: contentReversing(); // reversing the contents
51     break;
52     case 7: sortNumber(); // sorting the numbers
53     break;
54     case 8: countVowels(); // counting vowels
55     break;
56     case 9: findSpecificLine(); // finding desired line
57     break;
58     case 10: mergeFiles(); // merging two files
59     break;
60     case 0: cout << "Exiting..." << endl; // exiting the program
61     break;
62     default: cout << "Invalid choice. Please try again." << endl; // validating choice
63     break;
64 }
65
66 while (choice != 0); // condition for validating
67
68 return 0;
69 }
```

Task 1: Counting Words in a File

Write a C++ program that reads a text file and counts the number of words in the file.

Row (71-92)

```
71 void countWords() // function to count words in file
72 {
73     ifstream infile("Assignment_4.txt"); // reading from file
74
75     if(!infile) // handling error
76     {
77         cout<<"An error occurred in file opening. "<<endl;
78         return;
79     }
80
81     string word;
82     int word_count = 0;
83
84     while(infile >> word) // counting words
85     {
86         word_count++;
87     }
88
89     // displaying output
90     cout<<"\nTotal number of words in file 'Assignment_4' are: "<< word_count << endl;
91     infile.close(); // file closing
92 }
```

Task 2: Copying Contents from One File to Another

Write a C++ program that copies the contents of one file to another. Make sure to handle errors such as if the file does not exist.

Row (94-117)

```
94 void copyContents() // function to copy contents
95 {
96     ifstream infile("Assignment_4.txt"); // reading from file
97
98     if(!infile) // handle file opening error
99     {
100         cout<<"An error occured in file opening. "<<endl;
101         return;
102     }
103
104     ofstream outfile ("Copied From Assignment.txt"); // writing to new file
105
106     string word;
107
108     while(getline(infile,word)) // writing in file
109     {
110         outfile << word << endl;
111     }
112
113     // displaying the message
114     cout << "\nText copied from 'Assignment_4' to 'Copied From Assignment'. " << endl;
115     infile.close();
116     outfile.close(); // closing both files
117 }
```

Task 3: Finding Longest Word in a File

Write a C++ program that reads a file and finds the longest word in the file.

Row (119-145)

```
119 void longestWord() // function to find longest word
120 {
121     ifstream infile("Assignment_4.txt"); // reading from file
122
123     if(!infile) // handle file opening error
124     {
125         cout<<"An error occured in file opening. "<<endl;
126         return;
127     }
128
129     string word;
130     string longest_word;
131     int word_count = 0;
132
133     while(infile >> word) // compare the words
134     {
135         if (word.length() > longest_word.length())
136         {
137             longest_word = word;
138         }
139         word_count++;
140     }
141
142     // display the output
143     cout<<"The Longest Word in the File 'Assignment_4' is "<< longest_word << "." << endl;
144     infile.close(); // file closing
145 }
```

Task 4: Line Count in a File

Write a C++ program to count the number of lines in a file. Display the count as output.

Row (147-168)

```
147 void countLines() // function to count lines
148 {
149     ifstream infile ("Assignment_4.txt"); // reading from file
150
151     if (!infile) // handle file opening error
152     {
153         cout << "An error occurred in file opening. " << endl;
154         return;
155     }
156
157     string line;
158     int line_count = 0;
159
160     while (getline(infile, line)) // counting lines in file
161     {
162         line_count ++;
163     }
164
165     // display output
166     cout << "\nTotal Lines in the File 'Assignment_4' are: " << line_count << endl;
167     infile.close(); // closing file
168 }
```

Task 5: Replacing a Word in a File

Write a C++ program to search and replace a word in a file. The program should take input from the user for the word to be replaced and its replacement.

Row (170-208)

```
170 void replacingWord() // function to replace word
171 {
172     string word_replace, replacement_word; // local variables
173
174     cout << "Enter a Word to Replace: "; // taking input from user
175     cin >> word_replace;
176     cout << "Enter a Replacement Word: ";
177     cin >> replacement_word;
178
179     ifstream infile ("Assignment_4.txt"); // reading from file
180
181     if (!infile) // handle file opening error
182     {
183         cout << "An error occurred in file opening. " << endl;
184         return;
185     }
186
187     string content;
188     string line;
189
190     while (getline(infile, line)) // replacing the words
191     {
192         size_t pos = 0;
193         while ((pos = line.find(word_replace, pos)) != string::npos)
194         {
195             line.replace(pos, word_replace.length(), replacement_word);
196             pos += replacement_word.length();
197         }
198         content += line + "\n";
199     }
200     infile.close(); // closing file
201
202     ofstream outfile("Assignment_4.txt"); // writing to a file
203     outfile << content;
204     outfile.close(); // closing file
205
206     //display output
207     cout << "\nWord replaced successfully in file 'Assignment_4'." << endl;
208 }
```


Task 6: Reversing the Contents of a File

Write a C++ program that reads the contents of a file and writes them into another file in reverse order (character-wise).

Row (210-240)

```
210 void contentReversing() // function to reverse content of file
211 {
212     ifstream infile("Assignment_4.txt"); // reading from file
213     if (!infile) // handle file opening error
214     {
215         cout << "An error occurred in file opening. " << endl;
216         return;
217     }
218
219     string content;
220     string line;
221
222     while(getline(infile, line)) // reading
223     {
224         content = line + "\n" + content;
225     }
226     infile.close(); // close file
227
228     ofstream outfile ("Reversed_Assignment_4.txt"); // write to file
229     if(!outfile) // handle file opening error
230     {
231         cout << "An error occurred in file opening. " << endl;
232         return;
233     }
234
235     outfile << content; // write in file
236     outfile.close(); // file close
237
238     // display output
239     cout << "\nContents of file 'Assignment_4' reversed Successfully in 'Reversed_Assignment_4'. " << endl;
240 }
```

Task 7: Reading and Sorting Numbers from a File

Write a C++ program that reads a list of integers from a file, sorts them in ascending order, and writes the sorted numbers to another file.

Row (242-271)

```
242 void sortNumber() // function to sort numbers
243 {
244     ifstream infile ("Assignment_4Num.txt"); // reading from file
245     if(!infile) // handle file opening error
246     {
247         cout << "An error occurred in file opening. " << endl;
248         return;
249     }
250     int numbers[100]; // Assuming a maximum of 100 numbers
251     int count = 0;
252
253     while (infile >> numbers[count]) // reading number from file
254     {
255         count++;
256     }
257     infile.close(); // file close
258
259     // Simple bubble sort
260     for (int i = 0; i < count - 1; i++)
261     {
262         for (int j = 0; j < count - i - 1; j++)
263         {
264             if (numbers[j] > numbers[j + 1])
265             {
266                 int temp = numbers[j];
267                 numbers[j] = numbers[j + 1];
268                 numbers[j + 1] = temp;
269             }
270         }
271     }
```

Row (272-289)

```
272
273
274 ofstream outfile("Sorted_Numbers.txt"); // writing to file
275 if (!outfile) // handle file opening error
276 {
277     cout << "An error occurred in file opening." << endl;
278     return;
279 }
280 for (int i = 0; i < count; i++) // storing in file
281 {
282     outfile << numbers[i] << endl;
283 }
284 outfile.close(); // file closed
285
286 // display output
287 cout << "\nNumbers from file 'Assignment_4Num' sorted successfully in 'Sorted_Numbers'." << endl;
288 }
289
```

Task 8: Counting Vowels in a File

Write a C++ program that reads a file and counts the number of vowels (a, e, i, o, u) present in the file. Display the count for each vowel.

Row (290-322)

```
290 void countVowels() // function to count vowels
291 {
292     ifstream infile("Assignment_4.txt"); // reading from file
293     if (!infile) // handle file opening error
294     {
295         cout << "An error occurred in file opening." << endl;
296         return;
297     }
298
299     char vowels[] = {'a', 'e', 'i', 'o', 'u'};
300     int vowel_count[5] = {0}; // Count for each vowel
301
302     char ch;
303     while (infile.get(ch)) // comparing the words
304     {
305         ch = tolower(ch); // Convert to lowercase for comparison
306         for (int i = 0; i < 5; i++)
307         {
308             if (ch == vowels[i])
309             {
310                 vowel_count[i]++;
311             }
312         }
313     }
314     infile.close(); // file closed
315
316     // display output
317     cout << "\nFile 'Assignment_4' \nVowel counts:" << endl;
318     for (int i = 0; i < 5; i++)
319     {
320         cout << vowels[i] << ": " << vowel_count[i] << endl;
321     }
322 }
```

Task 9: Finding a Specific Line in a File

Write a C++ program that takes an integer *n* as input and prints the *n*th line of a file. If the line doesn't exist, print an appropriate message.

Row (324-354)

```
324 void findSpecificLine() // function to find specific line
325 {
326     int n; // taking line number from user
327     cout << "Enter the line number to retrieve: ";
328     cin >> n;
329
330     ifstream infile("Assignment_4.txt"); // reading from file
331     if (!infile) // handle file opening error
332     {
333         cout << "An error occurred in file opening." << endl;
334         return;
335     }
336
337     string line;
338     int line_number = 0;
339
340     while (getline(infile, line)) // finding the specific line
341     {
342         line_number++;
343         if (line_number == n)
344         {
345             cout << "\nFile 'Assignment_4'. \nLine " << n << ": " << line << endl;
346             infile.close(); // file close
347             return;
348         }
349     }
350     infile.close(); // file close
351
352     // display error
353     cout << "\nLine " << n << " does not exist." << endl;
354 }
```

Task 10: Merging Two Files

Write a C++ program to merge the contents of two text files into a third file. The program should ensure that the contents of both files are included sequentially.

Row (356-387)

```
356 void mergeFiles() // function to merge files
357 {
358     ifstream infile1("Assignment_4.txt"); // reading from file
359     ifstream infile2("Assignment_4Num.txt");
360     ofstream outfile("Merged_File.txt"); // writing to file
361
362     if (!infile1 || !infile2 || !outfile) // handle file opening error
363     {
364         cout << "An error occurred in file opening." << endl;
365         return;
366     }
367
368     string line;
369
370     // writing to file
371     while (getline(infile1, line))
372     {
373         outfile << line << endl;
374     }
375     while (getline(infile2, line))
376     {
377         outfile << line << endl;
378     }
379
380     // closing files
381     infile1.close();
382     infile2.close();
383     outfile.close();
384
385     // display output
386     cout << "\nFiles 'Assignment_4' and 'Assignment_4Num' are merged successfully." << endl;
387 }
```

Output

```
D:\University\Programming Fundamentals (Lab)\Assignments\Assignment 4\Assignment 4.exe

Menu:
1. Count Words
2. Copy Contents
3. Find Longest Word
4. Count Lines
5. Replace Word
6. Reverse File Contents
7. Sort Numbers
8. Count Vowels
9. Find Specific Line
10. Merge Files
0. Exit

Enter your choice: 1

Total number of words in file 'Assignment_4' are: 92

Menu:
1. Count Words
2. Copy Contents
3. Find Longest Word
4. Count Lines
5. Replace Word
6. Reverse File Contents
7. Sort Numbers
8. Count Vowels
9. Find Specific Line
10. Merge Files
0. Exit

Enter your choice: 2

Text copied from 'Assignment_4' to 'Copied From Assignment'.
```

```
Menu:
1. Count Words
2. Copy Contents
3. Find Longest Word
4. Count Lines
5. Replace Word
6. Reverse File Contents
7. Sort Numbers
8. Count Vowels
9. Find Specific Line
10. Merge Files
0. Exit

Enter your choice: 3

The Longest Word in the File 'Assignment_4' is vulnerabilitiee.

Menu:
1. Count Words
2. Copy Contents
3. Find Longest Word
4. Count Lines
5. Replace Word
6. Reverse File Contents
7. Sort Numbers
8. Count Vowels
9. Find Specific Line
10. Merge Files
0. Exit

Enter your choice: 4

Total Lines in the File 'Assignment_4' are: 10
```

```
Menu:
1. Count Words
2. Copy Contents
3. Find Longest Word
4. Count Lines
5. Replace Word
6. Reverse File Contents
7. Sort Numbers
8. Count Vowels
9. Find Specific Line
10. Merge Files
0. Exit

Enter your choice: 5
Enter a Word to Replace: s
Enter a Replacement Word: e

Word replaced successfully in file 'Assignment_4'.

Menu:
1. Count Words
2. Copy Contents
3. Find Longest Word
4. Count Lines
5. Replace Word
6. Reverse File Contents
7. Sort Numbers
8. Count Vowels
9. Find Specific Line
10. Merge Files
0. Exit

Enter your choice: 6

Contents of file 'Assignment_4' reversed Successfully in 'Reversed_Assignment_4'.
```

```
Menu:
1. Count Words
2. Copy Contents
3. Find Longest Word
4. Count Lines
5. Replace Word
6. Reverse File Contents
7. Sort Numbers
8. Count Vowels
9. Find Specific Line
10. Merge Files
0. Exit

Enter your choice: 7

Numbers from file 'Assignment_4Num' sorted successfully in 'Sorted_Numbers'.

Menu:
1. Count Words
2. Copy Contents
3. Find Longest Word
4. Count Lines
5. Replace Word
6. Reverse File Contents
7. Sort Numbers
8. Count Vowels
9. Find Specific Line
10. Merge Files
0. Exit

Enter your choice: 8

File 'Assignment_4'
Vowel counts:
a: 47
e: 78
i: 27
o: 33
u: 16
```

```
Menu:
1. Count Words
2. Copy Contents
3. Find Longest Word
4. Count Lines
5. Replace Word
6. Reverse File Contents
7. Sort Numbers
8. Count Vowels
9. Find Specific Line
10. Merge Files
0. Exit

Enter your choice: 9
Enter the line number to retrieve: 4

File 'Assignment_4'.
Line 4: the vulnerabilitiee or attack on the aeeete, each ae data theft or

Menu:
1. Count Words
2. Copy Contents
3. Find Longest Word
4. Count Lines
5. Replace Word
6. Reverse File Contents
7. Sort Numbers
8. Count Vowels
9. Find Specific Line
10. Merge Files
0. Exit

Enter your choice: 10

Files 'Assignment_4' and 'Assignment_4Num' are merged successfully.
```



```
Menu:
1. Count Words
2. Copy Contents
3. Find Longest Word
4. Count Lines
5. Replace Word
6. Reverse File Contents
7. Sort Numbers
8. Count Vowels
9. Find Specific Line
10. Merge Files
0. Exit

Enter your choice: 11
Invalid choice. Please try again.

Menu:
1. Count Words
2. Copy Contents
3. Find Longest Word
4. Count Lines
5. Replace Word
6. Reverse File Contents
7. Sort Numbers
8. Count Vowels
9. Find Specific Line
10. Merge Files
0. Exit

Enter your choice: 0
Exiting...
```

Thank You for Your Time!

The End